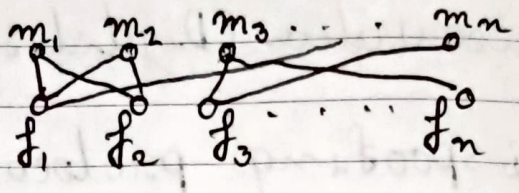
04/09/24

Advance Algorithms (PB)

Network flow

m = males

f = female



maximize # of matches : Stable mariage problem

Flow Network: → (Graph)

i) a directed graph $G = (V, E)$

ii) $(u, v) \in E$ has a non negetive capacity

$\{C(u,v) \geq 0 , C_e \geq 0\}$

iii) s (source) – no ingoing edge

t (sink) – no outgoing edge

iv) Every vertex lies on some path from source to sink. m = # edge, n = # vertice

$m \geq n-1$ [$\because$ < n-1 edges makes graph disconnected]

Flow: Given an s-t network, a flow (s-t flow) is a function f that maps each edge to non negetive real number & satisfies following two population.

i) capacity constraint: For all $(u,v) \in E$,

$f(u,v) \leq c(u,v)$

ii) Flow conservation (balance): For all $v \in V \setminus \{s,t\}$, the sum of flow along edge into v equals sum of flow along edges out of v.

If $(u,v) \notin E \Rightarrow f(u,v) = 0$

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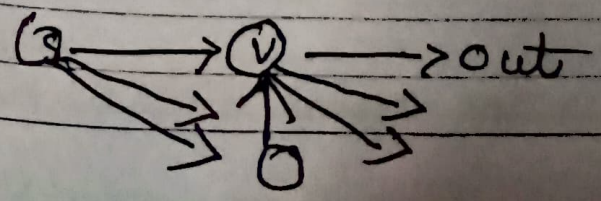
for $v \in V \setminus \{s,t\}$

$$f(v) = \sum_{(u,v) \in E} f^{in}(u,v) \text{ and } f^{out}(v) = \sum_{(u,w) \in E} f(u,w)$$



$$f^{in}(v) = f^{out}(v)$$

ii) flow conservation

$f(u,v)$: flow along edge (u,v)

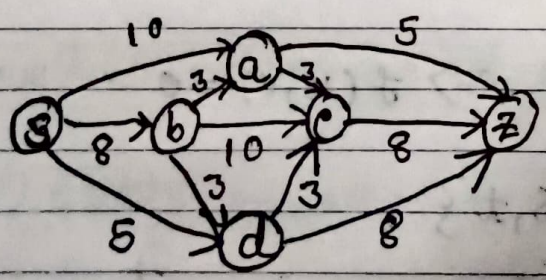
Total flow f: Amount of flow from s to t

Value of total flow $f = |f|$

$$|f| = f^{out}(s) = \sum_{w \in V} f(s,w) = f^{in}(t)$$

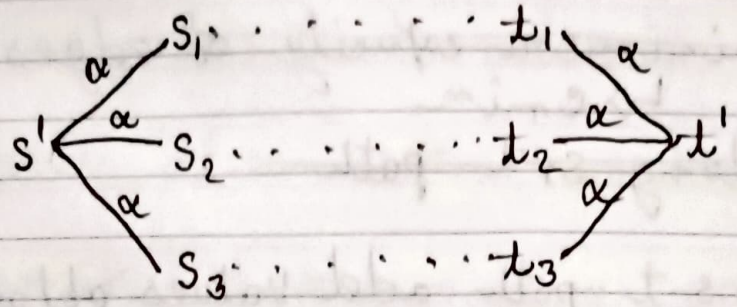
$$= \sum_{w \in V} f(w,t)$$

Maximum Flow: Given Flow network determine a flow of maximum value



$|f| = 21$ (total flow)

Multiple Source Multiple Sink



Edge based and path based flow

path based flow

① s 5 a 5 t -> 5

② s 3 a 3 c 3 t -> 3

③ s 5 b 5 c 5 t -> 5

④ s 3 b 3 d 3 z -> 3

⑤ s 5 d 5 t -> 5

21

Solution:

Greedy strategy

① Find an s-t path

② push minimum capacity of edges (u,v)

along the s-t path.

~~② push minimum capacity of edges~~
↳ c_{min}
~~(u,v) along s,t path~~

③ For all s-t path add values obtained in (2) to get the flow.

bottleneck capacity - min capacity on a path.

Ford-Fulkerson Algorithm

Residual network and augmenting path

→ mechanism to decrease flow in earlier chosen st path so that in current situation more flow can be pushed.

Given a flow network G and a flow f, the residual network G_f, is a network of some # of vertices & ~~some~~ source have following edges.

Forward edges: For each edge (u,v) for which $f(u,v) < c(u,v)$ create an edge (u,v) in G_f with $c(u,v) = c(u,v) - f(u,v)$

Backward edges: For each edge (u,v) for which $f(u,v) > 0$, create an edge (v,k) in G_f with capacity $c(v,k) = f(u,v)$

Ford_Fulkerson flow ($G=(V,E,s,t)$) {

$f = 0$ // all edges have 0 flow

$G =$ the residual network of G for f.

if G' has no s-t augmenting

path (augmenting path is an simple (no input of vertices) s-t path on G_f)

break;

P = any augmenting path of G'

c = minimum capacity of edge in path P augment f by adding c to the flow on every edge of P.

}

return f

}

① method of finding an s-t path

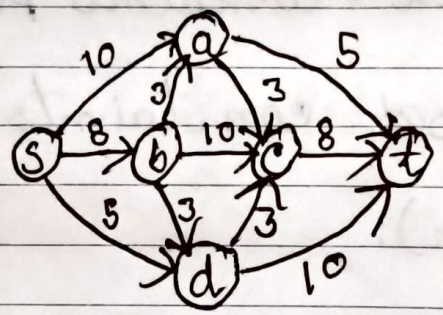
BFS/DFS → $O(|V|+|E|)$

$O(m+n)$

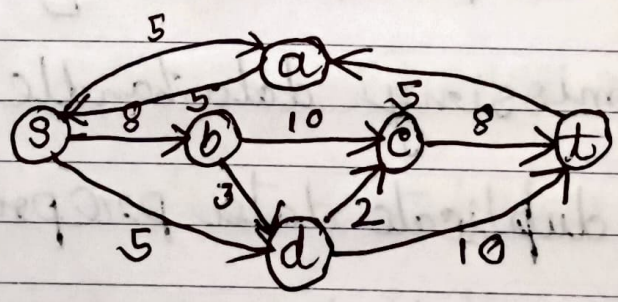
② How many time loop will be executed

③ : Does termination (no augmenting path)

given max flow



~~s-a-t~~

$\Rightarrow G_f$

s-a-t